# ML-based Cryptanalysis of Lightweight Ciphers

SIMON · SPECK · PRESENT

Evaluating the learnability of lightweight block ciphers using machine learning models — MLP, Logistic Regression, and CNN — across increasing rounds, measured by bit-level accuracy and Hamming distance.

## Instructor

Prof. Ravi Anand

## Course

Applied Cryptography

## Presented by

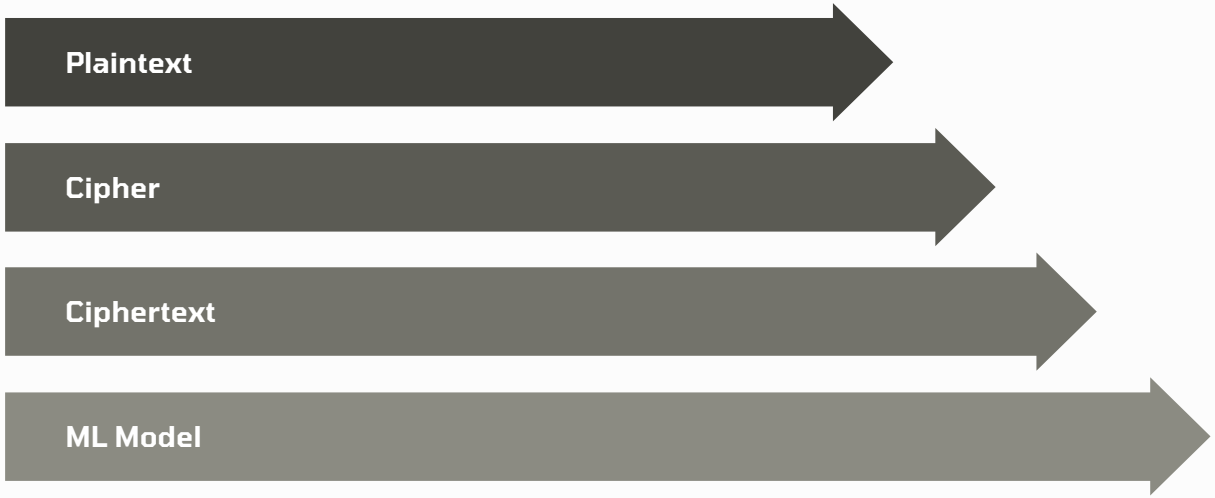
Nishant - Gautam - Areeb

# Problem Statement

## Core Research Question

**We treat encryption as a supervised learning problem.**

Can machine learning models learn to predict ciphertext directly from plaintext?




### Input

16-bit plaintext

### Output

16-bit ciphertext

### Task

Bitwise prediction: 16 independent binary classifiers

### Dataset

Automatically generated for reproducibility

METHODOLOGY

# Methodology

### 1 Plaintext

Original input data to be encrypted.

### 2 Cipher

Encrypted form derived from plaintext.

### 3 Dataset

Collected samples for training and testing.

### 4 ML Models

Sixteen independent models trained per bit.

### 5 Bit Predictions

Per-bit outputs produced by each model.

A step-by-step pipeline from data creation to bit-level prediction.

| Data | Rounds | Models |
| --- | --- | --- |
| 10k train / 2k test samples per round | 1 → 5 | 16 independent models, one per output bit |

Each output bit is learned independently — simplifies learning but reduces realism.

# Models Under Comparison

1

### Logistic Regression

A linear classifier used as a **baseline**. Can only capture linear separability between input and output bits. Expected to perform at chance (~50%) for most round configurations.

2

### MLP (Multilayer Perceptron)

A feedforward neural network capable of learning **non-linear relationships**. Multiple hidden layers allow it to approximate complex cipher transformations — particularly for low round counts.

3

### CNN

**Spatial pattern extractor.** Input reshaped to 4×4 grid before convolution — tests whether cipher output exhibits local spatial regularity.

# Evaluation Metrics

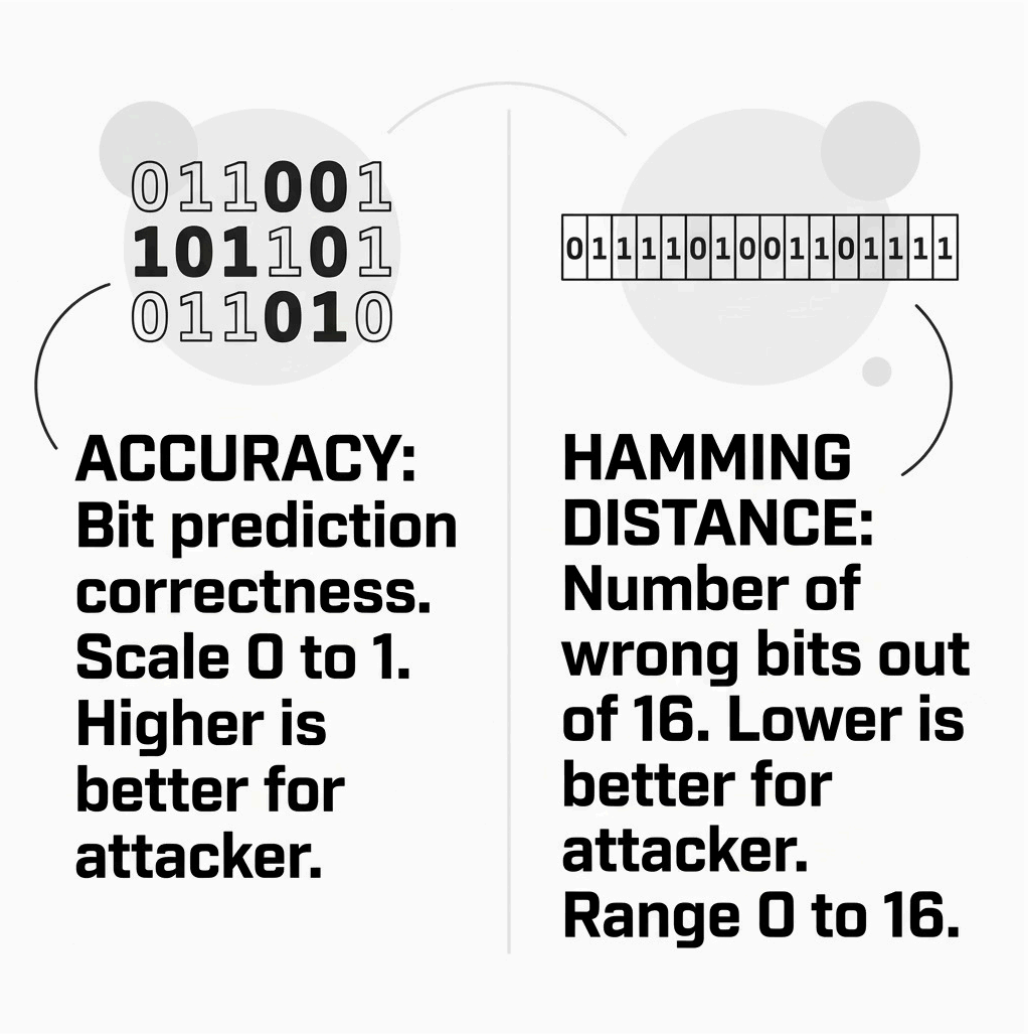
## Accuracy

Measures per-bit prediction correctness across all 16 binary classifiers. A value of 1.0 indicates perfect ciphertext reconstruction from plaintext alone.

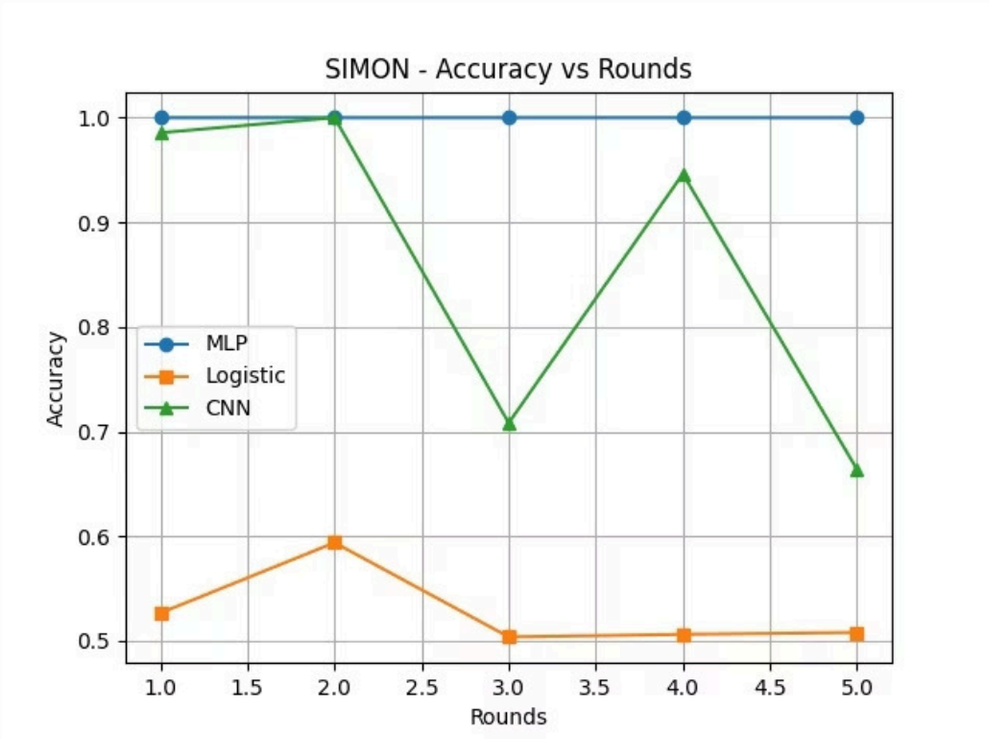
## Hamming Distance

Counts the number of differing bits between the predicted and true ciphertext. Captures *error magnitude* — not just whether the model is right or wrong, but by how much.

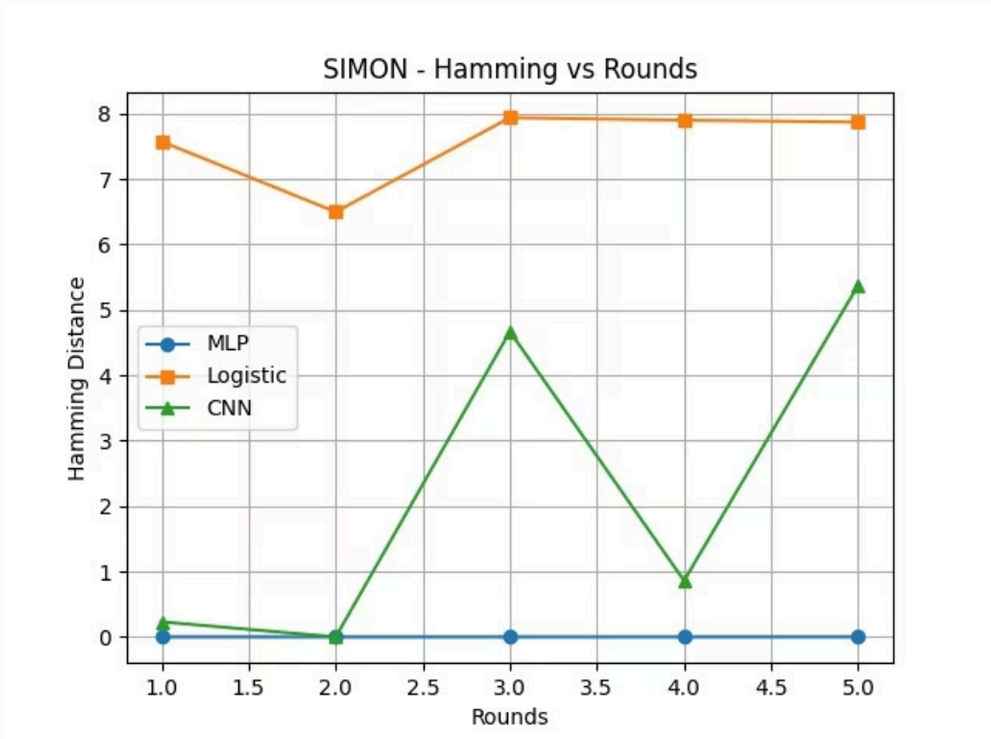
> **Key insight:** Lower Hamming distance = better cryptanalysis. A Hamming distance of 0 implies full plaintext-to-ciphertext learnability.



# SIMON Results




MLP achieves perfect accuracy. CNN shows dramatic instability, collapsing at rounds 3 & 5. Logistic Regression stays near 0.5 — no linear pattern.




Hamming distance reflects the accuracy trends — MLP near zero, CNN highly variable, Logistic Regression in the 7-8 range.

> **SIMON resists simple learning — CNN's collapse and Logistic's random-level performance indicate strong structural resistance to shallow models.**

# SIMON Analysis: Feistel Resistance & MLP Vulnerability

### MLP

Perfect accuracy (1.0) consistently — deep non-linear models can still memorize the full mapping despite Feistel structure

### CNN

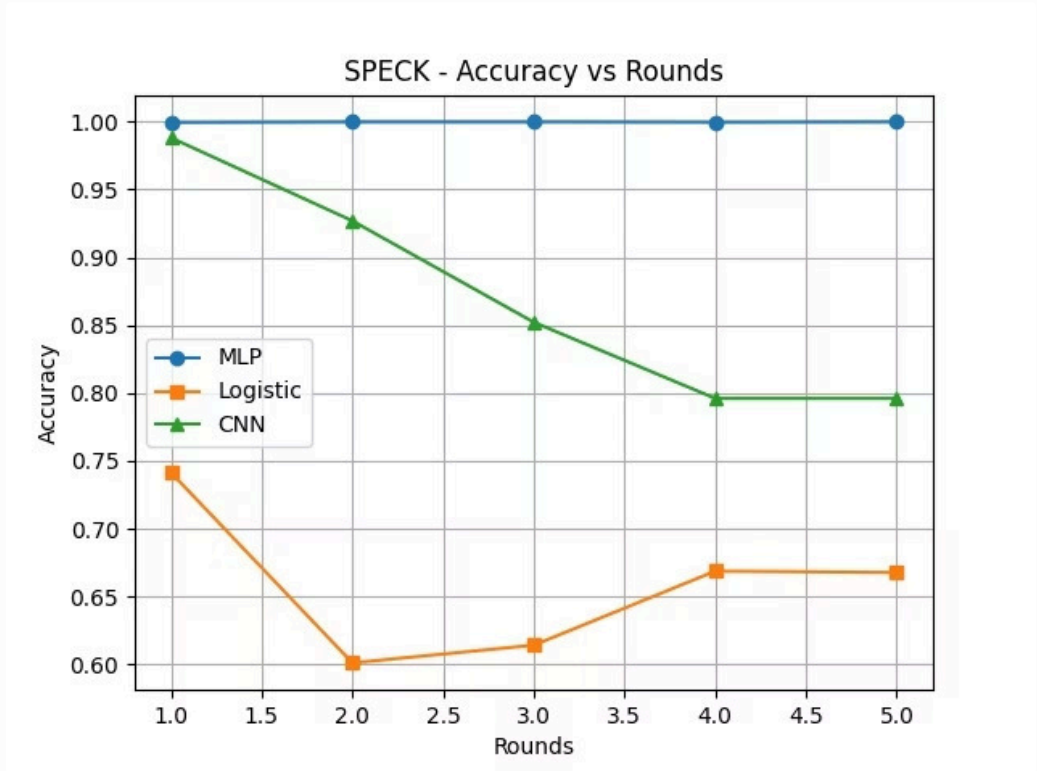
Highly unstable — collapses from 1.0 → 0.66 at rounds 3 & 5. Spatial patterns are inconsistent and unreliable

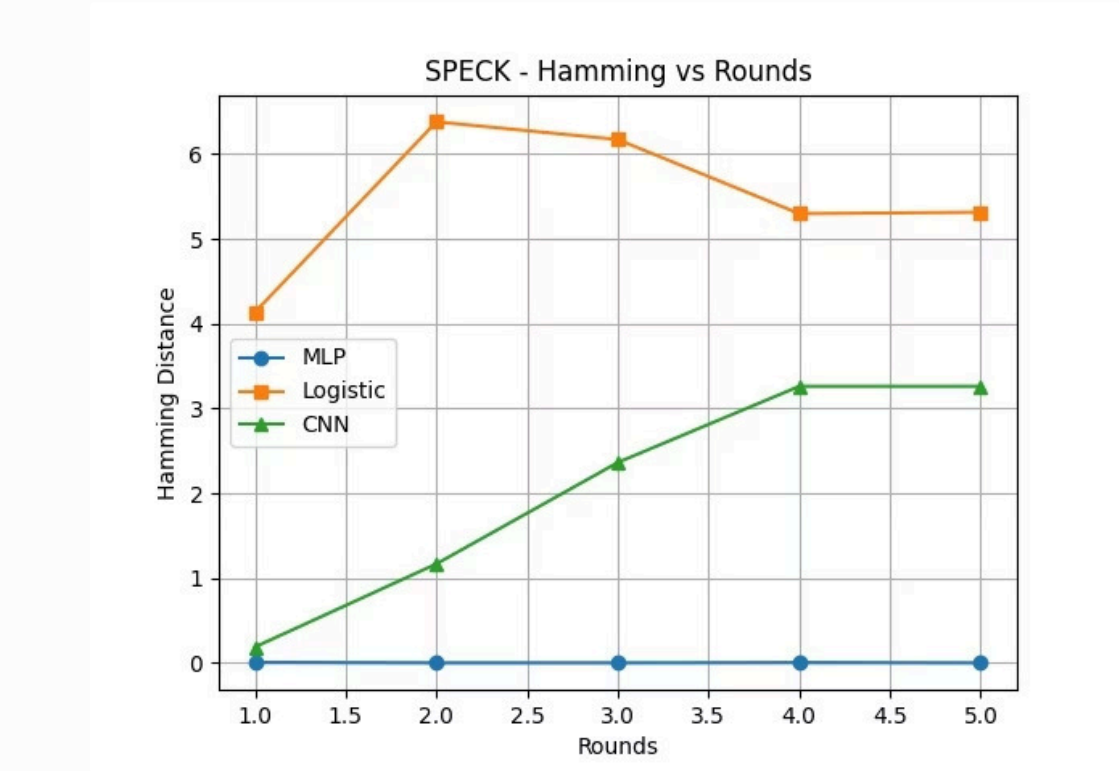
### Logistic Regression

Converges to ~0.5 — effectively random guessing. No linear separability despite Feistel's non-linearity

> **SIMON resists simple learning — CNN's instability and Logistic's random-level performance prove strong structural resistance. Yet MLP's perfect accuracy reveals that deep non-linear models can still break it through memorization rather than pattern learning.**

# SPECK Results




MLP maintains perfect accuracy (1.0). CNN drops steadily from 0.98 → 0.79. Logistic Regression ranges 0.6–0.7, showing weak linear separability.

Hamming distance reflects the accuracy trends — MLP near zero, CNN increases to ~3.2, Logistic Regression peaks at 6.2.

SPECK's arithmetic complexity (ARX operations) progressively defeats pattern-based models as rounds increase.

# SPECK Analysis: ARX Complexity & Strongest Resistance

### MLP

Perfect accuracy (1.0) consistently — memorization persists despite ARX complexity

### CNN

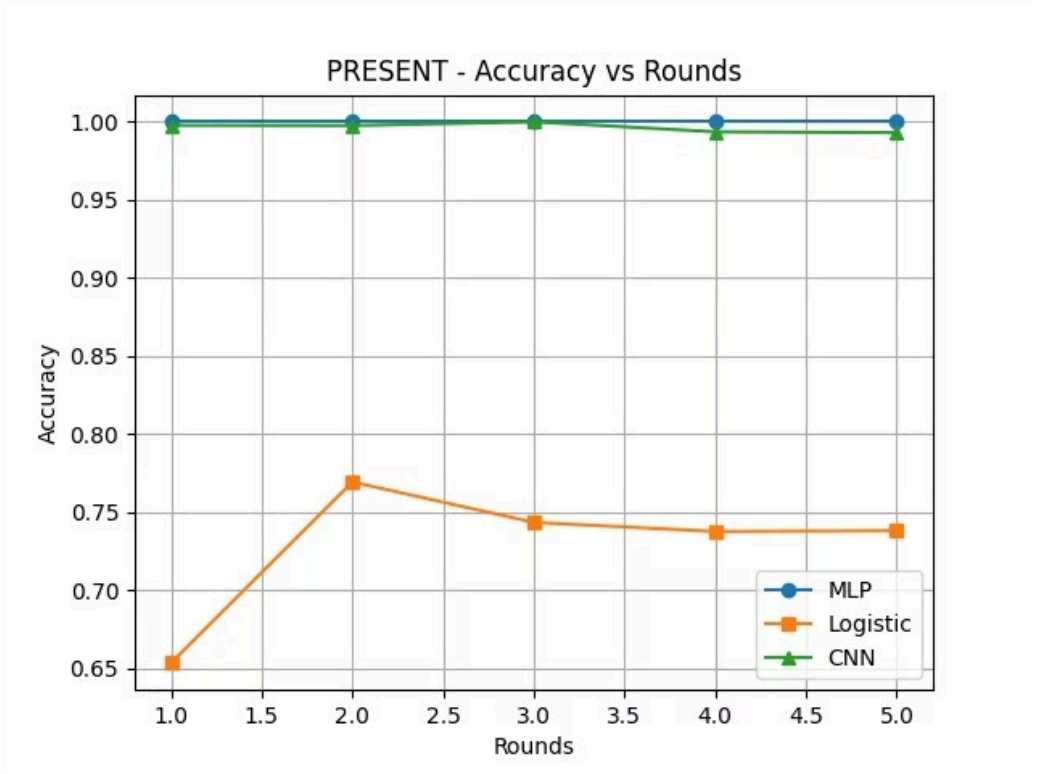
Steady degradation from 0.98 → 0.79 — arithmetic operations dissolve spatial patterns progressively

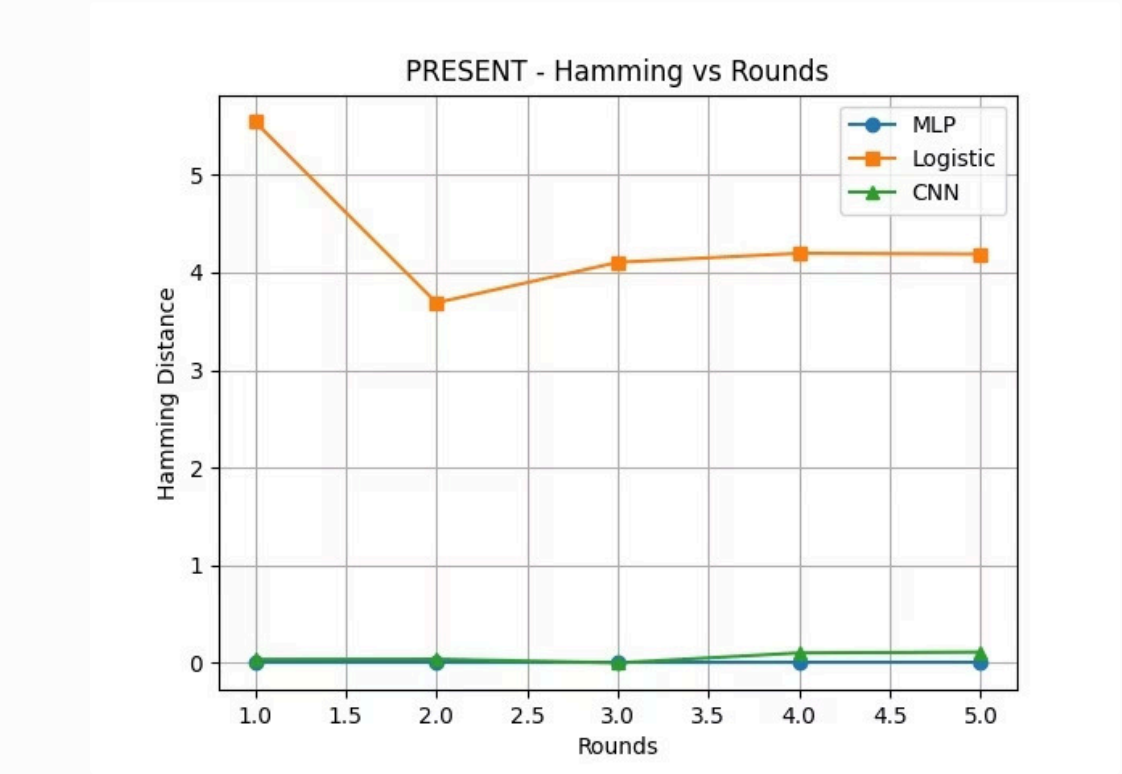
### Logistic Regression

Weak, ranging 0.6–0.7 — ARX construction resists linear separation effectively

> **SPECK demonstrates the strongest resistance among the three ciphers. Arithmetic operations with rotation create non-local dependencies that defeat CNN's spatial assumptions entirely. Yet MLP's perfect accuracy reveals that deep memorization still breaks it — indicating that full-round SPECK requires additional diffusion layers for cryptographic security against neural networks.**

# PRESENT Results




Accuracy remains consistently high across rounds 1-5, with MLP at 1.0, CNN near 0.99, and Logistic Regression around 0.74.

Hamming distance stays near zero for MLP and CNN, while Logistic Regression remains in the 4-6 range across the five rounds.

The graphs highlight PRESENT's strong learnability and the clear gap between deep models and the linear baseline.

# PRESENT Analysis: Weak Diffusion Exposed

### MLP

~100% accuracy across all rounds, near-zero Hamming distance

### CNN

~99% accuracy, strong spatial regularity detected

### Logistic Regression

~0.74, above-chance indicating partial linear separability

> **If ML predicts ciphertext with near-perfect accuracy, the cipher's internal structure is being directly exposed. Even shallow models break PRESENT at low round counts — a strong signal of insufficient diffusion.**

# Hamming Distance Insights

**0**

**MLP**

Perfect reconstruction, full plaintext-to-ciphertext mapping learned

**4–8**

**Logistic Regression**

Random-like guessing, high error across all ciphers

**~2–5**

**CNN**

Partial structure learning, medium error indicating incomplete mapping

Hamming distance confirms accuracy trends clearly. MLP's distance of 0 is the most alarming finding — it means the model outputs the exact ciphertext from plaintext alone, making key recovery a tractable next step. Logistic Regression's 4–8 bit errors are equivalent to near-random guessing. CNN's medium error shows it captures partial structural patterns but fails to learn the complete mapping.

# Final Comparison: ML Security View

### PRESENT — Weakest

Broken by all three models. Even Logistic Regression achieves above-chance accuracy. Insufficient diffusion at low rounds.

### SIMON — Moderate

Resists linear and spatial models. Non-linear Feistel structure adds complexity — but MLP still memorizes the mapping perfectly.

### SPECK — Strongest (Relative)

ARX operations degrade CNN performance significantly. Highest effective resistance among the three — yet still MLP-learnable.

→ **PRESENT:**

High learnability → weak diffusion

→ **SIMON:**

Resists linear models, but MLP memorizes

→ **SPECK:**

Strongest due to ARX, but still MLP-learnable

**Critical takeaway:** All three ciphers are broken under MLP at low round counts. This indicates overfitting to insufficient diffusion — not a flaw in cipher design at full rounds, but a clear signal that round-reduced variants expose exploitable structure.

# Limitations of This Study

To ensure a focused analysis and enable the use of Machine Learning models, several constraints were applied to our experimental setup. These limitations are crucial for interpreting the results accurately and understanding the scope of our findings.

### Small Block Size

All ciphers were evaluated with a **16-bit block size** for computational feasibility, not their standard operational sizes.

### Low Rounds

Analysis was restricted to **1–5 rounds**, targeting the initial diffusion properties rather than full-round security.

### Fixed Key

A **fixed key** was used throughout, allowing models to learn the plaintext-ciphertext mapping for a specific key, not general key recovery.

### Bitwise Independent Learning

Models predict each output bit independently, simplifying the learning task but not reflecting a true ciphertext reconstruction.

**Important Context:** This research focuses on *structural analysis*, not a practical attack. It **does NOT break real-world cipher security** for full-round ciphers.

# Conclusion

Our study has illuminated critical aspects of lightweight cipher security through the lens of machine learning, revealing both vulnerabilities and strengths across different designs.

### ML Can Exploit Patterns

Even shallow neural networks can identify subtle, exploitable patterns within round-reduced ciphers, suggesting structural weaknesses.

### Increased Rounds for Security

Higher round counts consistently correlate with increased resistance against ML attacks, highlighting the importance of sufficient diffusion.

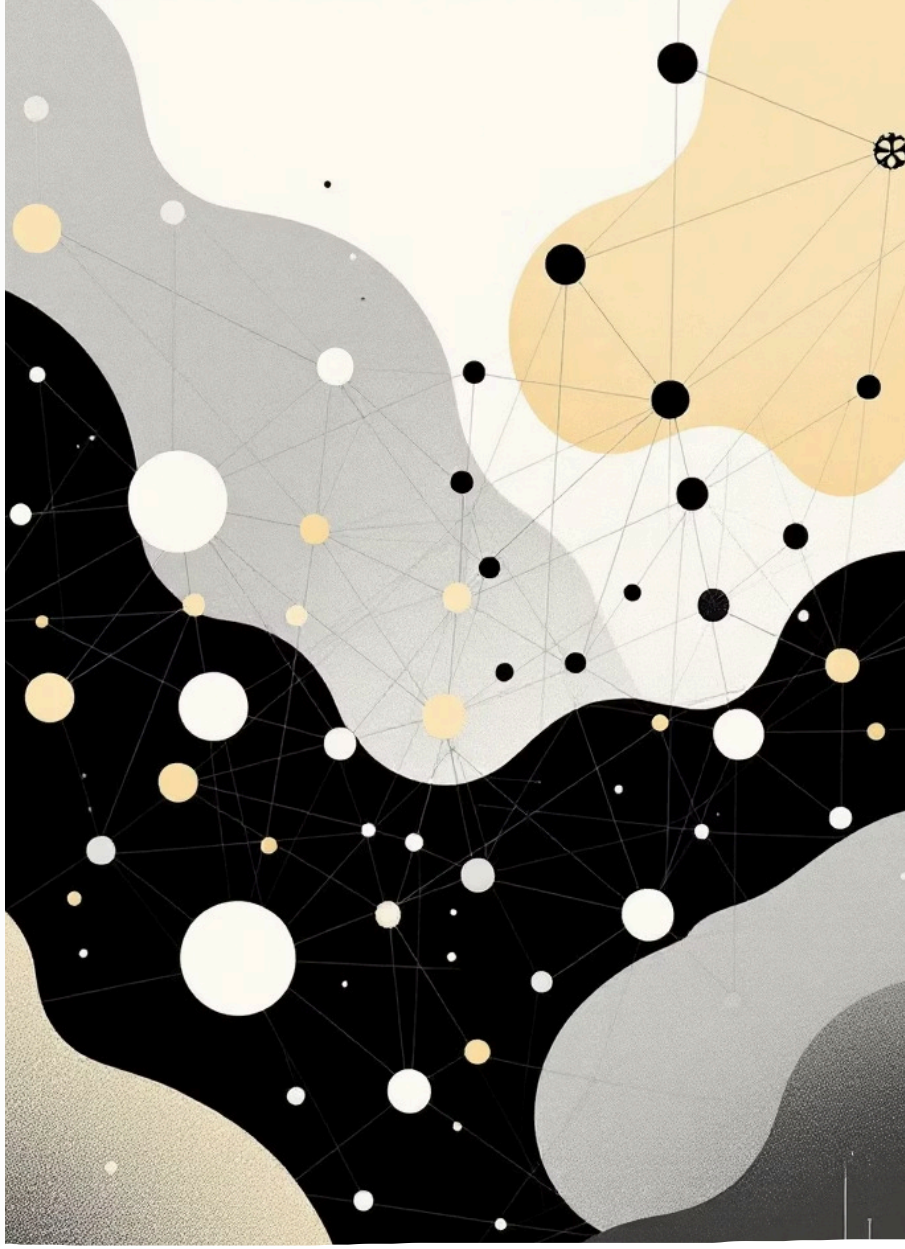
### The Ideal: Appears Random

A cryptographically strong cipher should effectively hide its internal operations, appearing indistinguishable from random noise to any adversary, including ML models.

> **A secure cipher should appear random even to neural networks.**

# Thank You

We appreciate your attention and engagement throughout this presentation.

## Questions?

Please feel free to ask any questions you may have regarding our study or findings.